

Chapter 1

Section 6 - Counting

1.1 Section 6.1 - Intro to Counting

There are 2 highways from Palm Springs to Upland and 3 highways from Upland to Corona. How many ways can one travel from palm Springs to Corona if you must travel through Upland

	PU1	PU2
UC1		
UC2		
UC3		

$$2 \cdot 3 = 6 \text{ ways}$$

This is the fundamental theorem of Counting

Theorem 1 (Fundamental Theorem of Counting)

If there are a ways to do task A and b was of doing task B, then there are $a \cdot b$ ways to do task A and B

Example 1

Let's do some example of this

1. How many 2 letter initials are there if...
 - (a) The letters can be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{26}_{\text{Letter 2}} = \boxed{676}$$

(b) The letters CANNOT be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{25}_{\text{Letter 2}} = \boxed{650}$$

Since we can choose any letter for the first one, but the last one we can choose all but the one we just chose, so $26 - 1 = 25$

2. How many 7 digit phone numbers do not start with 0 or 1?

$$\underline{8} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = \boxed{8,000,000}$$

3. How many ways can 7 people line up for a picture?

(a) There are no restrictions

$$\underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 7! = \boxed{5040}$$

(b) Dina must be the first person in the photo

$$\underline{1} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6! = \boxed{720}$$

(c) Dina is NOT the first person in the photo

$$\underline{6} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6 \cdot 6! = \boxed{4320}$$

OR

ways Dina w/ first + # ways Dina NOT first = total

ways Dina NOT first = total - # ways Dina w/ first

$$= 5040 - 720$$

$$= \boxed{4320}$$

(d) Dina must stand next to Omar

We do this by treating Dina and Omar as one pair

But since there's two ways of arranging Dina and Omar, Dina - Omar and Omar - Dina, we gotta multiply it every twice

$$2(\underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1}) = 2 \cdot 6! = \boxed{1440}$$

(e) Dina must not stand next to Omar

$$\begin{aligned}
 \# \text{ Dina next to Omar} + \# \text{ Dina NOT next to Omar} &= \text{total} \\
 \# \text{ Dina NOT next to Omar} &= \text{total} - \# \text{ Dina next to Omar} \\
 &= 5040 - 1440 \\
 &= \boxed{3600}
 \end{aligned}$$

4. A bit string is a sequence of 0's and 1's. How many bit strings are there of length 7?

(a) If there are no restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^7 = \boxed{128}$$

(b) Begin with 01

$$\underline{1} \cdot \underline{1} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^5 = \boxed{32}$$

We could also just say how many ways can we arrange a 5 length bit string since the first two are fixed (01)

(c) Are palindromes, ie are the same when written forwards or backwards

$$\underbrace{\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2}}_{\text{anything}} \cdot \underbrace{\underline{1} \cdot \underline{1} \cdot \underline{1}}_{\text{fixed}} = 2^4 = \boxed{16}$$

5. How many ways can 5 people order a single scoops of icecream if the possible flavors are vanilla, coffee, chocolate, strawberry, or olive?

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

Since there's 5 flavors and 5 people

6. How many whole numbers less than 100 are...

(a) Divisible by 3

$$\left\lfloor \frac{100}{3} \right\rfloor = 33 \# \text{ of possible ints div by 3}$$

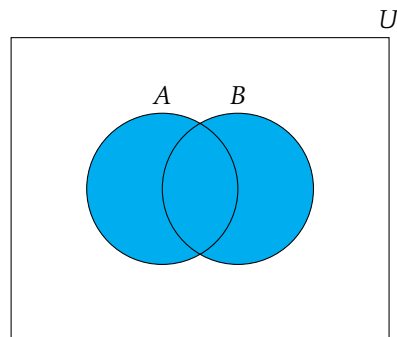
$$\left\lfloor \frac{100}{3} \right\rfloor + \underbrace{1}_{\text{including 0}} = \boxed{34}$$

(b) div by 7?

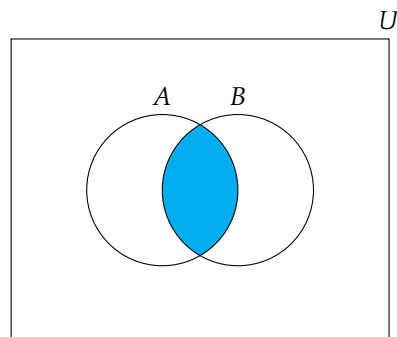
$$\left\lfloor \frac{100}{7} \right\rfloor + 1 = 14 + 1 = \boxed{15}$$

(c) div by 3 OR 7?

This represents 3 OR 7, where A is all numbers div by 3 and B is same but for 7



This represents div by BOTH 3 and 7



Recall 1

$$A \cup B = A + B - A \cap B$$

Being divisible by both 3 and 7 just means being divisible by 21

$$A \cap B = \left\lfloor \frac{100}{3 \cdot 7} \right\rfloor + 1 = 4 + 1 = 5$$

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 34 + 15 - 5 \\ &= \boxed{44} \end{aligned}$$

7. A die is rolled 5 times. How many ways can this happen if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} \cdot \underline{6} \cdot \underline{6} \cdot \underline{6} = 6^5 = \boxed{7776}$$

(b) 4 is never rolled

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

(c) The number 4 is rolled at least once

ways 4 is rolled at least once + # never rolled = total

ways 4 is rolled at least once = total - # never rolled

$$= 7776 - 3125$$

$$= 6^5 - 5^5$$

$$= \boxed{4651}$$

8. A pair of dice is rolled. How many outcomes are possible if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} = \boxed{36}$$

(b) The sum is 4

$$(1, 3), (3, 1), (2, 2)$$

so 3 ways

9. A coin is flipped 10 times. How many outcomes are possible if

(a) No restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

(b) At least 1 of the flips is heads

$$\begin{aligned} \# \text{ ways no heads} + \# \text{ ways at least 1} &= \text{total} \\ \# \text{ at least 1} &= \text{total} - \underbrace{\# \text{ ways no heads}}_{\text{all tails}} \\ &= 2^{10} - 1 \\ &= \boxed{1023} \end{aligned}$$

1.1.1 Multiples of n up to k

Theorem 2 (Number of Multiples of n up to k)

To find the number of positive integers divisible by n up to k is given by

$$\left\lfloor \frac{k}{n} \right\rfloor$$

This DOES NOT include 0, so for all whole numbers it would be

$$\left\lfloor \frac{k}{n} \right\rfloor + 1$$

1.2 Section 6.2 - Pigeonhole Principle

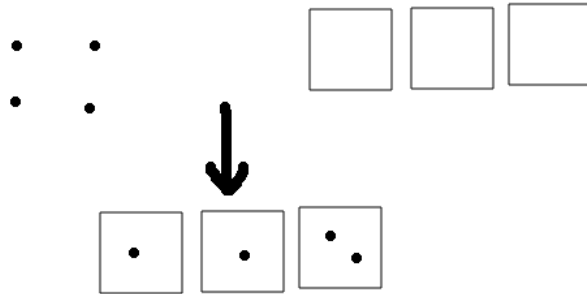
Definition 1 (Pigeonhole Principle)

If there are n boxes and more than n objects placed in these boxes, at least one box contains more than one object.

At least one box will contain

$$\left\lceil \frac{k}{n} \right\rceil$$

objects



Example 2

There are 43 people in this room. Are we guaranteed that

1. At least 2 people have the same birth month?

Solution:

Yes, since 12 months ; 43 people

1.3 Section 6.3 - Permutations and Combinations

Let's start with permutations

1.3.1 Permutations

Suppose you have 8 different books. How many ways can the books on a shelf be arranged if...

1. The shelf fits all 8 books?

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 8! = \boxed{40320}$$

2. The shelf fits 5 books

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = \frac{8!}{3!} = \frac{8!}{(8-5)!} = \boxed{6720}$$

Here, the order matters. We call this a **permutation**

Definition 2 (Permutation)

A **Permutation** is one specific way a group of items can be arranged where

order matters. In general, if we have n objects and we want to arrange k of them, then the number of ways is given by

$${}_nP_k = \frac{n!}{(n-k)!}$$

Example 3

How many ways can the president, vice president, and treasurer be assigned from a group of 4 people?

$$\underbrace{4}_P \cdot \underbrace{3}_{VP} \cdot \underbrace{2}_T = \boxed{24}$$

or, using the formula

$${}_4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \boxed{24}$$

Now let's move on to Combinations

1.3.2 Combinations

Example 4

How many committees of 3 people can be formed from 4 people?

Here, order **DOES NOT** matters

Here, there are 4 different committees

ABC	ABD	ACD	BCD
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You can arrange the 3 people 6 different ways, given by

$${}_3P_3 = 3! = 6$$

Note

I would write everything out but I'm too lazy

Another way of getting that there are 4 committees is

$$\frac{\text{total \# ordered arrangements}}{\text{\# ways of permuting that arrangement}} = \frac{24}{6} = \frac{\frac{4!}{(4-3)!}}{3!} = \frac{4!}{3!(4-3)!}$$

This is the number of **Combinations**

Definition 3 (Combination)

A **Combinaton** is just an arrangement where order does not matter.

The number of combinations of k items taken from n items is

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

You would say "n choose k"

Think of this as "how many ways can you arrange k things in n slots"

Example 5 1. How many ways can 2 cards be dealt from a standard deck of cards?

52 cards, want 2 cards

$$\begin{aligned}\binom{52}{2} &= \frac{52!}{2!52-2!} \\ &= \frac{52!}{2!50!} \\ &= \frac{52 \cdot 51}{2} \\ &= \boxed{1326}\end{aligned}$$

2. A coin is flipped 10 times. How many ways can exactly 4 of the flips be heads?

$$\begin{aligned}\binom{10}{4} &= \frac{10!}{4!6!} \\ &= \boxed{210}\end{aligned}$$

notice how this also works for 6 flips to be tails!

3. Ten distinct points are labeled on a circle. How many quadrilaterals can be drawn using these points?

$$\binom{10}{4} = \frac{10!}{4!6!} = \boxed{210}$$

There's a LOT more examples but I'm too lazy to write them all out. Just go back to the notes and write them out later

Let's prove that $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

Proof

$$\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!}$$

=

uhhh do later (it's on the very back of 6.3)

1.4 Section 6.4 - Binomial Identities

$$\begin{aligned} (a+b)^0 &= 1 \\ (a+b)^1 &= 1a + 1b \\ (a+b)^2 &= 1a^2 + 2a^1b^1 + 1b^2 \\ (a+b)^3 &= 1a^3 + 3a^2b^1 + 3a^1b^2 + 1b^3 \\ (a+b)^4 &= 1a^4 + 4a^3b^1 + 6a^2b^2 + 4a^1b^3 + 1b^4 \\ (a+b)^5 &= 1a^5 + 5a^4b^1 + 10a^3b^2 + 10a^2b^3 + 5a^1b^4 + 1b^5 \\ &\vdots \\ (a+b)^{10} &= 1a^{10} + \underbrace{10}_{\binom{10}{9}} a^9b^1 + \underbrace{45}_{\binom{10}{8}} a^8b^2 + \underbrace{120}_{\binom{10}{7}} a^7b^3 + \underbrace{210}_{\binom{10}{6}} a^6b^4 + \underbrace{252}_{\binom{10}{5}} a^5b^5 + \underbrace{210}_{\binom{10}{4}} a^4b^6 + 120a^3b^7 + 45a^2b^8 + \dots \end{aligned}$$

For all of these, it's 10 choose either the power of a or the power of b . This is because the power is essentially the number of arrangements of $aaabbbbbbb$ for example

This brings us to our next theorem

1.4.1 Binomial Theorem

Theorem 3 (Binomial Theorem)

$$\text{For } n = 0, 1, 2, \dots, (a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

I would put the proof here but it's long and I'm lazy

Example 6 1. A pizza parlor has 10 toppings. How many pizzas are possible if any number of toppings can be chosen but none can be repeated?

$$\begin{aligned} & \underbrace{\# 0 \text{ toppings}}_{\binom{10}{0}} + \underbrace{\# 1 \text{ topping}}_{\binom{10}{1}} + \dots + \underbrace{\# 10 \text{ toppings}}_{\binom{10}{10}} \\ &= \sum_{k=0}^{10} \binom{10}{k} \underbrace{1^{10-k} \cdot 1^k}_{\text{to use B.T}} \\ &= (1 + 1)^{10} \\ &= 2^{10} \\ &= \boxed{1024} \end{aligned}$$

2. How many bit strings are there of length 10?

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

or

Place k 0's in the bit string.

$$\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \dots + \binom{10}{10} = (1 + 1)^{10} = 2^{10} = 1024$$

3. How many subsets does a set with 10 elements have?

uhhhh same thing

1.4.2 Sum of Combinations

Theorem 4 (Sum of Combinations)

$$\text{For } n = 0, 1, 2, \dots, \sum_{k=0}^n \binom{n}{k} = 2^n$$

Example 7

uhhhhh just use the formula

1.5 Section 6.5 - Stars and Bars